FRP frame profiles: vendors and visit brief for HSW

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The next build's frame moves to pultruded FRP profiles in vinyl ester resin, fire-retardant grade, joined with FRP clips that are both bolted and bonded. Sanchuang in Shipai, Dongguan, is the closest real pultrusion factory to Shenzhen and the first one to visit. Exel Composites in Nanjing is the engineering-heavy option for joint design. Shenzhen itself has no structural pultruders we could find. Most of China's pultrusion industry is in Hebei and Jiangsu.

Vendors

Company details come from each vendor's own listing and are unverified. Confirm resin options before booking a visit.

Vendor	Where	Why this one	Check first
Sanchuang FRP Products Factory 东莞市石排三创玻璃纤维制品厂	Chikan Village, Shipai Town, Dongguan. About 1 to 1.5 h by road from Shenzhen	Makes the shapes we need in-house: angle, U-channel, square and round tube, H profiles, flat strip. Founded 2006, 51 to 100 staff, 10,000 to 30,000 m² of factory, ISO 9001	Resin systems aren't published, so ask for vinyl ester and a fire-retardant grade. The listed contact is Louis Liu, through the SupplierGO page
Exel Composites (Nanjing)	No. 2120 Chengxin Dadao, Jiangning Science Park, Nanjing	Finnish pultrusion maker with two plants in Nanjing. Offers vinyl ester profiles for chemical resistance. The best-placed of the three to advise on joint design	A flight from Shenzhen, so start with a video call. Ask about minimums and pricing for a 20-unit run

Vendor	Where	Why this one	Check first
Qingyuan Puhe New Material Technology 清远普和新材料 (Made-in-China listing)	Qingyuan, Guangdong. About 2.5 h by road from Shenzhen	Pultrusion specialist, ISO 9001 and ISO 14001 per its listing	Its catalogue is mostly rod, flat bar and stakes. Confirm it makes structural channel and angle before going

Dongguan Guding (东莞市固鼎建材) advertises 800 t a year of pultruded products, but its 2023 registry entry shows four staff at an office address, so it is probably a trader. Skip it.

Visit brief

Bring the frame layout with the full unit weight and where the tank, cells and casters sit. Explain the application in these terms:

- Frame for an outdoor pool chlorine generator. Saturated salt brine can splash or leak inside the cabinet, and chlorine gas can reach the frame.
- Installed outdoors in hot sun, with parts inside the cabinet reaching about 70 °C.
- Small runs. About 20 units next, ideally as kits that arrive cut to length and drilled.
- Joints are FRP clip angles or gusset plates, bolted and bonded. No metal fasteners where brine can reach.

Questions to put to the vendor:

1. Do you make these profiles in vinyl ester resin, in a fire-retardant grade? Which fire test does it pass (UL 94 or a GB standard), and can we have the report? Do the profiles have a UV-resistant surface veil?
2. Which standard channel, angle, square tube and flat plate sizes do you stock that suit a frame this size? What does a custom die cost?
3. What corner joint do you recommend? What minimum edge distance and bolt spacing suit your wall thicknesses?
4. Which structural adhesive do you use on your profiles, methacrylate or epoxy? What surface prep does it need, and how strong is it at 70 °C?
5. Do you supply FRP bolts and nuts (玻璃钢螺栓)? In what sizes, and with what torque limit?
6. Do you seal cut ends and drilled holes with resin? Will you supply kits cut, drilled and sealed?

7. Can you give us mechanical properties for the profiles, including at 70 °C, and chemical resistance to salt brine and hypochlorite?
8. What are the minimum order and lead time for a 20-unit kit? Can you make a sample corner joint for us to soak in hot brine and load-test before we commit?

Sources

- Sanchuang FRP Products Factory, SupplierGO company page and its profiles listing
- Exel Composites, standard structural profiles and Exel's Nanjing expansion
- Made-in-China, FRP pultruded profile suppliers
- Dongguan Guding, Baidu Baike entry
- Strongwell Design Manual, Section 19 (connections)